

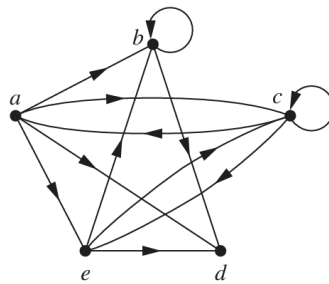
Leading University
Department of Computer Science and Engineering
CSE – 1151 (Discrete Mathematics)
Quiz Exam

Full Marks: 10 (2 x 5)

[Answer any 2 questions]

Time: 20 mins

1. The message "L ORYH PB FRXQWUB YHUB PXFK" was encrypted using a Caesar cipher. Decrypt the message to find the original text.
2. Using the Fast Modular Exponentiation method, find $8^{222} \bmod 11$.
3. Find the in-degree and out-degree of each vertex in the graph G with directed edges.



Leading University
Department of Computer Science and Engineering
CSE – 1151 (Discrete Mathematics)
Quiz Exam

Full Marks: 10 (2 x 5)

[Answer any 2 questions]

Time: 20 mins

1. Encrypt the plaintext message "EXPERIENCE IS A GREAT TEACHER" using the shift cipher with shift key $k = 23$.
2. Using the Fast Modular Exponentiation method, find $7^{225} \bmod 11$.
3. Using Dijkstra's algorithm, find the shortest path and the length between a and z of the following weighted graph.

